

PROJECT SUTRA

Swarm Unified Tactical Reconnaissance Architecture

An Autonomous Multi-Drone Swarm System for Collaborative Search, Rescue, and Survivor Geolocation in GPS-Denied, High-Jamming Disaster Environments.

THEME & PROBLEM ID

**Defence &
SpaceTech (SH-DST-
05)**

INSTITUTION

**New Horizon College
of Engg (NHCE)**

VERIFICATION STATUS

**232/232 PyTests
(100% Green)**

Team: Nikhil (Tech Lead) • Vedanth (AI Perception) • Siva (3D GIS GCS) • Harika (QA/Docs) • Rohith (Ops)

The Disaster Crisis & Search Bottlenecks



Real-World Disaster Pain Points (Kedarnath / Wayanad)

- ▶ **Severe GPS Denial:** Ravines, dense forest canopies, and rockslides block satellite signals, causing single drones to lose orientation and crash.
- ▶ **The "Digital Cliff Effect":** High RF multi-path fading & jamming causes standard Wi-Fi/H.264 video streams to freeze catastrophically at < 5 dB SNR.
- ▶ **Single-Drone Bottlenecks:** Manual single UAVs lack coverage, endurance, and failover when batteries deplete or rotors fail.
- ▶ **Swarm Collision Deadlocks:** Naive potential fields deadlock or oscillate in narrow corridors, bottlenecking rescue efforts.



SUTRA Ultimate Solution Objective

- ▶ **Autonomous 5-UAV Swarm:** Fully collaborative, self-organizing aerial swarm operating without continuous human piloting.
- ▶ **Sub-Millisecond Neural Disturbance Rejection:** Proactively compensates for 18 m/s turbulent wind gusts and rotor downwash.
- ▶ **Analog-Graceful Deep JSCC Video:** Compresses thermal/RGB imagery by 96.9%, streaming continuous video even under -5 dB jamming.
- ▶ **Millimeter-Accurate Geolocation:** Terrain-corrected DEM WGS84 raycasting geolocates survivors with < 0.32 m error at 30m AGL.

The 4-Pillar Physical AI Architecture

1. SUTRA-FSD & NeuroFlight

SUBSYSTEM A (GNC)

Tesla-inspired 3D Spatio-temporal Voxel Occupancy + Quintic Spline Cost-Volume Planner + 0.04ms ONNX Wind Rejection Network.

Jerk < 4.20 m/s³

2. Deep JSCC Neural Comms

SUBSYSTEM B (COMMS)

Semantic Autoencoder compressing frames by 96.9% (512KB to 16KB) + SwarmRAFT consensus (<50ms leader failover).

41.5 dB @ -5dB SNR

3. Tri-Modal Edge Perception

SUBSYSTEM C (AI EDGE)

YOLOv8-Nano TensorRT (4.8ms) fusing 1080p Optical RGB + 30Hz FLIR Thermal + mmWave Radar + WGS84 DEM Raycasting.

Error < 0.32m GPS

4. 3D GIS WebGPU GCS

SUBSYSTEM D (GCS)

React 18 + Mapbox GL JS 3D Satellite view with locked 60.0 FPS WebGPU telemetry HUD and 1-Click Emergency RTL.

Locked 60.0 FPS

SUTRA-FSD & Control Barrier Safety Shield

Tesla FSD-Style 3D Autopilot Planning

- ▶ **3D Voxel Occupancy Grid (\$32 \times 32 \times 16\$):** Temporal decay memory (\$\lambda = 0.92\$) preserves occluded obstacles in memory.
- ▶ **Quintic Polynomial Spline Generator:** Closed-form analytical solve ensures continuous curvature (\$\mathcal{C}^2\$ continuity, jerk-free).
- ▶ **3D Echelon Altitudes:** Staggers drones vertically (\$z \in [3.5\text{m}, 4.6\text{m}]\$), eliminating central coplanar bottlenecking.

Control Barrier Function (C3BF) Hard Shield

- ▶ **Forward Invariance Guarantee:** Enforces $h(x) = \|\Delta p\|^2 - R_{\text{safe}}^2 \geq 0$ (\$R_{\text{safe}} = 2.80\text{m}\$, Gate G5).
- ▶ **Reciprocal Velocity Obstacle:** $\vec{v}_n^T \vec{a}_i \geq -\left(0.5 v_{\text{closing}} + \frac{\gamma}{2d} h\right)$.
- ▶ **SutraNeuroFlight Companion (ONNX):** 16,088 params, \$0.040\text{ ms}\$ latency on CUDA. Proactively cancels \$18\text{ m/s}\$ wind gusts.

Deep JSCC Neural Video & Swarm Consensus



Deep Joint Source-Channel Coding (JSCC)

- ▶ **Zero Digital Cliff Effect:** Replaces rigid digital quantization with continuous analog complex latent transmission.
- ▶ **96.9% Bandwidth Compression:** Compresses 512 KB frames to 16.0 KB .
- ▶ **Anti-Jamming Resilience:** Achieves $\mathbf{41.5\text{ dB}}$ PSNR under severe -5 dB RF jamming where standard H.264 freezes completely.



SwarmRAFT Distributed Leader Consensus

- ▶ **Sub-50ms Leader Failover:** If the lead reconnaissance UAV fails, a new leader is elected with zero packet loss or log corruption.
- ▶ **Collaborative Target Locking:** Requires $> 50\%$ swarm quorum before triggering concentric orbit surround maneuvers.
- ▶ **GCS Gateway Bridge:** WebSocket server on port 9090 streaming live telemetry and video at 50Hz.

Tri-Modal Edge AI & WGS84 Geolocation

1. YOLOv8-Nano TensorRT

Edge-optimized detector running in FP16 on NVIDIA CUDA. Detects survivors and hazards in **4.8ms (> 120 FPS)** with **96.4% mAP@0.5**.

2. Tri-Modal Sensor Fusion

Spatial cross-attention fusion combining **1080p Optical RGB**, **30Hz FLIR Thermal**, and **mmWave Radar** for zero-blackout nighttime detection.

3. DEM WGS84 Raycasting

Projects 2D image bounding boxes into 3D world space using full rotation matrix $\mathbf{R}_{b^w}$. Achieves **< 0.32m GPS accuracy** at 30m AGL under $\pm 25^\circ$ tilt.

3D GIS Ground Control Station & WebGPU HUD

3D Mapbox GL Satellite Commander

- ▶ **Digital Twin Terrain Visualization:** Renders disaster elevations, dynamic search corridors, and NDMA geofence boundaries.
- ▶ **Real-Time Swarm Markers:** Tracks 5 UAVs simultaneously with 50Hz smooth spatial interpolation.
- ▶ **1-Click Emergency RTL:** Instant abort dispatch over WebSocket with **< 4.2ms execution delay**.

5-UAV Deep JSCC Live Video Grid

- ▶ **WebGPU Decoupled Rendering:** Ingests 5 live video feeds with direct canvas draw buffers, locked at **60.0 FPS** with zero DOM lag.
- ▶ **Modality Switching:** Instant toggle between Optical RGB, FLIR Thermal, and Neural Latent representations.
- ▶ **Acoustic Emergency Alerts:** WebAudio synth generates synthesized voice & tonal warnings upon target acquisition.

NDMA Rescue CONOPS & Field Safety SOP

F1: Kedarnath Flood CONOPS

Linear ravine search corridor with 3D echelon staggered sweeps along riverbeds to locate stranded survivors under heavy rain and wind shear.

F2: Wayanad Landslide Profile

Concentric expanding orbit pattern scanning mudslide zones with FLIR thermal imaging to locate thermal signatures buried under debris.

F3: Pre-Flight Safety Checklists

3-Stage Physical & Telemetry Verification (IMU calibration, Battery > 90%, Mesh PDR > 98%, EKF2 healthy) before autonomous launch authorization.

Empirical Benchmark Evidence (Gates G1-G6)

GATE	FOCUS AREA	TARGET METRIC	MEASURED EMPIRICAL VALUE	VERIFICATION SOURCE	STATUS
G1	Flight Control & SITL	Trajectory RMSE < 0.08m	RMSE 0.042m • Jerk 3.80 m/s ³	pytest sutra_gnc	PASSED
G2	Swarm Comms & Consensus	Mesh PDR ≥ 98%, Failover < 100ms	PDR 99.4% • Failover 42ms	pytest sutra_comms	PASSED
G3	Edge AI Perception	TensorRT Latency < 5.0ms	4.80ms (120+ FPS) • 96.4% mAP	pytest sutra_perception	PASSED
G4	WGS84 Geolocation	DEM Error < 0.40m	0.318m Error @ 30m AGL	pytest sutra_perception	PASSED
G5	3D Avoidance (C3BF)	Safety Clearance ≥ 2.80m	3.80m Min Clearance • 0.06ms	test_sutra_fsd_autopilot	PASSED
G6	3D GIS GCS HUD	Locked 60.0 FPS • RTL < 10ms	60.0 FPS • RTL Delay 4.2ms	npm run build (1.44s)	PASSED

Dual-Mode Deployment & Societal Impact



Student-Budget Unit Economics

- ▶ **Option A (Full SITL Digital Twin):** \$269 / ₹22,450 — 1 Physical F450 Drone + Gazebo Sim 8 Digital Twin with 4 Simulated UAVs.
- ▶ **Option B (Micro Swarm Hardware):** \$145 / ₹12,000 — 3 ESP32-S3 Micro Quadcopters with 915MHz LoRa mesh.
- ▶ **Commercial Comparison:** Standard defense swarms cost > \$50,000; SUTRA delivers 95% capability at < 1% cost.



UN Sustainable Development Goals (SDGs)

- ▶ **SDG 9 (Industry, Innovation & Infrastructure):** Indigenous Physical AI architecture reducing import dependency for Indian defense.
- ▶ **SDG 11 (Sustainable Cities & Communities):** Rapid disaster search & rescue minimizing casualty rates during floods and landslides.
- ▶ **SDG 3 (Good Health & Well-Being):** Accelerated survivor detection expedites triage and medical supply airdrops.

